

- 4 The circulatory system and the cardiac cycle ensure that all the cells of the body are supplied with their requirements.

(a) The table shows the diameter of the lumens of an aorta and a capillary, and the thickness of their walls.

Type of blood vessel	Diameter of lumen	Thickness of wall / mm
aorta	2.5 cm	2.0
capillary	0.14 μm	0.001

(i) How many times greater is the diameter of the lumen of the aorta than the diameter of the lumen of the capillary?

(1)

- ☐ A 17 800
- ☐ B 17 900
- ☐ C 178 000
- ☐ D 179 000

(ii) Explain why the wall of the aorta is thicker than the wall of a capillary.

(3)

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- (b) The table shows the durations of ventricular diastole and ventricular systole at two different heart rates.

Heart rate / beats per minute	Duration of ventricular diastole / seconds	Duration of ventricular systole / seconds
75	0.53	0.27
200	0.14	0.16

Determine the relationship between heart rate and the duration of ventricular diastole and ventricular systole.

(2)

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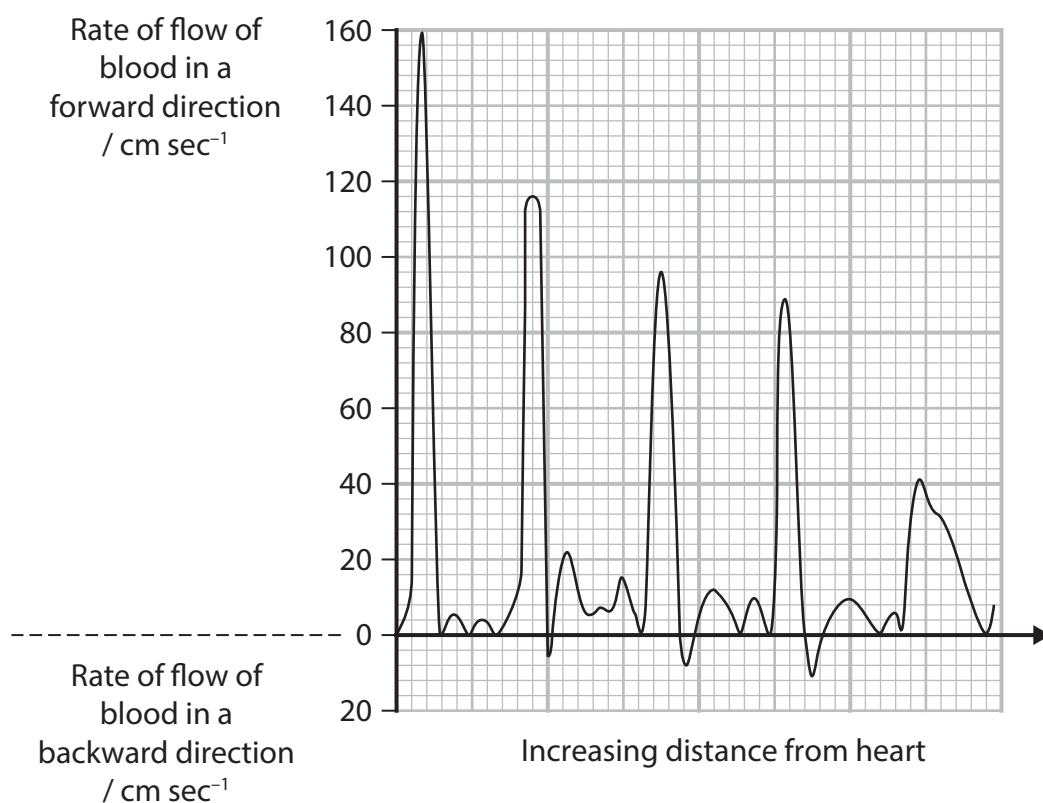
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- (c) The graph shows the rate of flow of blood through the aorta of a dog with increasing distance from the heart.



Explain the changes in rate and direction of flow of blood through this aorta.

Use the information in the graph to support your answer.

(3)

(Total for Question 4 = 9 marks)

